

Bingying Wang

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Education

University of Utah

Aug. 2021 - May 2026(Expected)

Degree: Bachelor | **Major:** Computer Science, **Minor:** Mathematics | **GPA:** 3.84/4.0

Scholarships & Honors:

Academic Excellence Scholarship

Apr. 2021 - May 2025

Dean's List

May 2022, Dec 2022, Dec 2024, May 2025

Technical Skills

Programming Languages: C(1 yr), C#(3 yrs), C++(1 yr), Java(4 yrs), JavaScript(1 yr), Python(4 yrs), R(2 yrs), Ruby, Racket

Technical Skills: Machine Learning (Python), Data Analytics (SQL, MATLAB), Frontend Development (HTML, CSS, JavaScript), AI Model Development, Physics Engines (Unity, QT)

Practical Operation Tools: Linux, Git, Docker, Visual Studio Code, WordPress, JetBrains Suite (IntelliJ IDEA, PyCharm)

Languages: English (Proficient), Mandarin (Native), Japanese (Intermediate)

Selected Projects

Reinforcement Learning for Robotic Manipulation

LL4MA Lab, Kahlert School of Computing, UT

Research Assistant Supervised by Associate Professor Tucker Hermans

Jan. 2025 - May 2025

- Designed and implemented a custom Gymnasium environment for the Franka Panda robotic arm in PyBullet to perform pick-and-place manipulation tasks.
- Trained a Soft Actor-Critic (SAC) policy to learn grasping and lifting behaviors with dense reward shaping and domain randomization for improved generalization.
- Exploring hierarchical control by separating grasping and placement policies, coordinated through a higher-level switching policy.

Center for High Performance Computing (CHPC) Security Project

Kahlert School of Computing, UT

Research Assistant Supervised by Associate Professor Jun Xu

May 2024 - Aug. 2024

- Designed user surveys and conducted data analysis to inform the development of robust security measures, approved by the Institutional Review Board (IRB).
- Developed Python-based data analysis workflows to clean and extract user log data from the CHPC system and combine it with SQL for data filtering and statistical aggregation.

Internship Experience

University of Maryland College Park

College Park, MD

Undergraduate Research Intern, Vision-Language-Action (VLA) in Robotics

Jan. 2026 – Present

- Conducted a structured literature review on VLA datasets and benchmarks; developed a taxonomy across data source, embodiment diversity, modalities, and action-space formulations.
- Built and maintained a comparative dataset table (real vs. synthetic) and wrote the “VLA Datasets” section, emphasizing design trade-offs, reproducibility, and evaluation implications.
- Collaborating with graduate students and faculty mentors to iterate on model design, run ablation studies, and prepare research results for internal reports/paper drafts.

Tongji Architectural Design (Group) Co., Ltd.

Shanghai, China

Software Engineering Intern, Engineering Department

May 2025 - Aug. 2025

- Participated in Shanghai's urban infrastructure project and developed back-end services in Python for a cloud-based urban planning platform to improve system scalability and performance.
- Designed and implemented RESTful APIs to integrate real-time municipal infrastructure data, optimizing data processing and visualization.

University of Utah

Salt Lake City, UT

Teaching Assistant for CS1420, CS 2420 and CS3390

Aug. 2023 - May 2024, Jan. 2026 - Present

- Guided more than 300 students in mastering *Object-Oriented Programming* (CS1420), *Data structures and Algorithms* (CS2420), and *Ethics in Data Science* (CS3390).
- Supported professors with assignment grading and exam solution preparation to ensure accurate and consistent grading.
- Facilitated discussion-based activities and case studies, supporting students' ethical reasoning across the data analysis and AI pipeline.
- Collaborated with faculty to address student learning challenges and improve overall academic performance.